BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Storage Units and Conversions

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What should guide the choice of a storage device?

2. When does a compiler translate source code?
   A) When a key is pressed
   B) Only after the program ends
   C) Before the program runs
   D) Only while each line runs

3. For scores = [5, 1, 3], total starts at 0 and each element is added once. Give the successive totals and the final result.

4. Use decimal units. Convert 5 TB to GB, then state how many TB are in 1 PB.

Today's objectives

Recall bit, nibble, byte and decimal storage-capacity units. Convert correctly between bits, bytes, kB, MB and GB.

Key vocabulary - Read, cover, recall

Bit
A binary digit, either 0 or 1.

Nibble
Four bits.

Byte
Eight bits.

Kilobyte
1000 bytes using decimal SI units.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Small units

A bit stores one binary digit; a nibble is four bits and a byte is eight bits.

Larger units

For GCSE storage-capacity questions here: 1 kB = 1000 bytes; 1 MB = 1000 kB; 1 GB = 1000 MB. State assumptions where relevant.

Conversion direction

Moving to a smaller unit multiplies; moving to a larger unit divides. Bits and bytes require a factor of eight.

Capacity calculations

Convert all sizes to the same unit before dividing total capacity by file size.

I Do - Worked model

Convert 4 MB into kB and then into bytes using decimal units. [4 marks]

- State the conversion factor
- 1 MB = 1000 kB and 1 kB = 1000 bytes.
- 4 x 1000 = 4000 kB.
- 4000 x 1000 = 4,000,000 bytes. [4]

We Do - Guided practice

1. Convert 3 GB into MB and kB. Show both steps. [4]

2. Convert 56 bits to bytes and explain the operation. [3]

3. A 5 GB drive stores files of 250 MB each. Calculate how many fit. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Display the number of leftover bits when 43 bits are grouped into bytes.

```
bits = 43
leftover = ...
print(leftover)
```

You Do - Independent exam questions

Q2. Convert 2.5 MB into kB using decimal units. [2]

Q3. Calculate the number of bits in 12 bytes. [2]

Q4. A device has 6 GB free. Each file uses 750 MB. Calculate the maximum whole files it can store. [3]

Exit Ticket - True or False?

An algorithm only gives reliable results when its inputs are handled as specified.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. The application's requirements, including capacity, speed, cost and physical use.

2. C - Before the program runs

3. 5, 6, 9; final total 9.

4. (a) 5,000 GB; 1 PB = 1,000 TB.

I Do - Explain the model

State the conversion factor

1 MB = 1000 kB and 1 kB = 1000 bytes.

4 x 1000 = 4000 kB.

4000 x 1000 = 4,000,000 bytes. [4]

We Do - Model answers

1. 3 x 1000 = 3000 MB [2]; 3000 x 1000 = 3,000,000 kB [2].

2. There are 8 bits in a byte [1]; 56 / 8 [1] = 7 bytes [1].

3. 5 GB = 5000 MB [1]; 5000 / 250 [1] = 20 files [1], assuming no overhead [1].

Code First - Answer

```
bits = 43
leftover = bits MOD 8
print(leftover)
```

Expected output: 3

You Do - Self-mark

Q2: 2.5 x 1000 [1] = 2500 kB [1].

Q3: 12 x 8 [1] = 96 bits [1].

Q4: 6 GB = 6000 MB [1]; 6000 / 750 [1] = 8 whole files [1], ignoring overhead.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.